(Slot - c)

$CO_2$ - $AT_1$

Computer Vision - ITA0502

Analytical Problem Solving

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38/50

1) Grey level Transformation :-

(a) · It enhances image quality by modifying pixel intensities.
· It improves brightness, contrast, and visibility of image details.
· It is commonly used for preprocessing before segmentation and feature extraction.

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(b) Negative Transformation :-

(formula :-) $S = L - 1 - r$

→ For an 8-bit image :- $S = 255 - r$

how?

Eg :-

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| Original (r) | Negative (s) |
|---|---|
| 0 | 255 |
| 50 | 205 |
| 100 | 155 |
| 150 | 105 |
| 200 | 55 |
| 255 | 0 |

Bright regions become dark and dark regions become bright.

2) Histogram Equalization :-

Purpose : Histogram equalization improves image contrast by redistributing intensity values over the available range.

PDF ?

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CDF :-

| Intensity | Frequency | Cumulative | CDF |
|---|---|---|---|
| 0 | 4 | 4 | 4/26 = 0.154 |
| 1 | 6 | 10 | 10/26 = 0.385 |
| 2 | 10 | 20 | 20/26 = 0.769 |
| 3 | 6 | 26 | 26/26 = 1.000 |

(c) Equalized Intensities : For a 4-bit image $L=4$ :

$S=(L-1)\times CDF$

| Original | New Value. |
|---|---|
| 0 | 0 |
| 1 | 1 |
| 2 | 2 |
| 3 | 3 |

Interpretation : The histogram becomes more evenly distributed, improving contrast.

3) Gaussian Smoothing :-

(a) Why Gaussian Filter :-
- Reduces random noise effectively
- Produces smoother images than an averaging filter
- Gives higher weight to center pixels
- Preserves edges better.

(b) Center Pixel Calculation :-

$$\frac{1}{16}\begin{bmatrix}1 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 1\end{bmatrix}$$

Multiply each kernel value with the corresponding image pixel, add them and divide by 16.

Interpretation :- Noise is reduced while maintaining important image structures.

4) Edge Detection Using Sobel Operator :

(a) purpose : • Detects objects boundaries.
- Finds intensity changes.
- Used in segmentation, feature extraction.

(b) Horizontal Sobel Mask :-

$$\begin{bmatrix} -1 & -2 & 1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

• Large Positive / Negative Value → Strong horizontal edge.

• Value near zero → No significant edge.

5) Contrast Stretching :-

(a) Need :-

Contrast stretching expands narrow intensity range to the full display range, improving visibility and making objects easier to distinguish.

(b) Formula :-

For mapping :- $[50, 150] \rightarrow [0, 255]$

$$S = \frac{(r - 50) \times 255}{150 - 50}.$$

Example :-

| Original | New Value |
|---|---|
| 50 | 0 |
| 75 | 64 |
| 100 | 128 |
| 125 | 191 |
| 150 | 255 |

Interpretation: The image uses the full grayscale range, giving higher contrast and better visual quality.